State Monad

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Purely thread mutable state.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Game score threaded through pure moves.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.state;

import java.util.function.Function;

public record State<S, A>(Function<S, Result<S, A>> run) {
    public record Result<S, A>(S state, A value) {}

    public Result<S, A> run(S s) { return run.apply(s); }

    public <B> State<S, B> map(Function<? super A, ? extends B> f) {
        return new State<>(s -> {
            var r = run(s);
            return new Result<>(r.state(), f.apply(r.value()));
        });
    }

    public <B> State<S, B> flatMap(Function<? super A, State<S, B>> f) {
        return new State<>(s -> {
            var r = run(s);
            return f.apply(r.value()).run(r.state());
        });
    }

    public static <S, A> State<S, A> pure(A a) { return new State<>(s -> new Result<>(s, a)); }
    public static <S> State<S, S> get() { return new State<>(s -> new Result<>(s, s)); }
    public static <S> State<S, Void> set(S s) { return new State<>(ignored -> new Result<>(s, null)); }
    public static <S> State<S, Void> modify(Function<S, S> f) {
        return new State<>(s -> new Result<>(f.apply(s), null));
    }
```

Classic Implementation (2)

```
}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/state/
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick State Monad? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Purely thread mutable state.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Game score threaded through pure moves. [0:40] Intent — State the design force in one sentence: Pure...